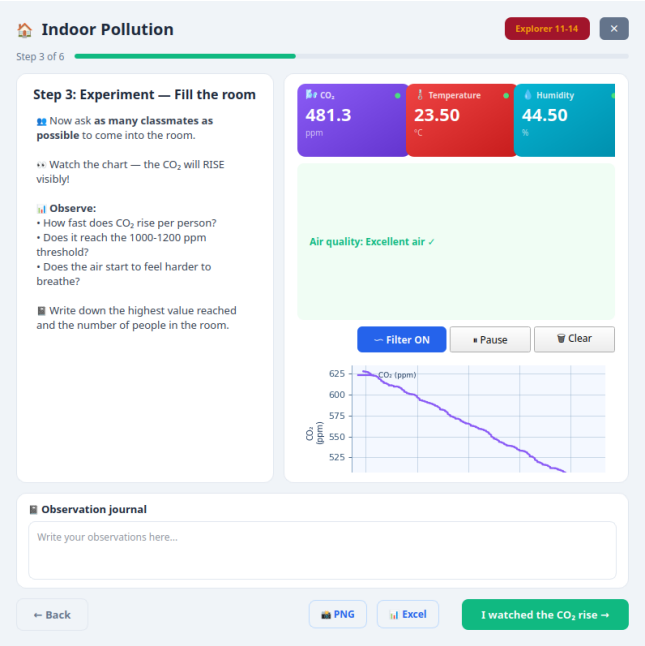
Indoor Pollution

Your own classroom's accumulation curve — and a ventilation recommendation backed by data

Level: Explorer — ages 11–15 (Romanian school years V–VIII) • **Time in class:** 20–25 min, one 50-minute lesson, or a project with a full logging day • **Sensors:** SCD41 (carbon dioxide, temperature, humidity)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer — ages 11–15 (Romanian school years V–VIII). The most direct entry is in **year VII** (biology: human respiration), and the most productive mathematically in **year VI**, where ratios make the central operation possible: **the rate of rise**, in ppm per minute. In year V the whole activity can be done by comparing values and curve shapes, without computing any rates.

Assumed knowledge

Reading a graph over time and subtracting two values. For rates: **the ratio of two differences** — ppm divided by minutes — so from year VI. For the project: the arithmetic mean and a chart in a spreadsheet.

Time in class

Short: 20–25 min. Full: one 50-minute lesson, with all four measurements written down and the rates computed. Project: three 50-minute lessons (2 h 30 min) over about a week, plus **one unattended logging day**, which costs no class time.

Teacher preparation

15 min — power up the box and leave the SCD41 for ten minutes to check that the readings look sensible: outdoors or at an open window it should settle near 400–450 ppm. **Decide and mark where the box will stand** — roughly at the head height of a seated pupil, away from the window, away from the door, and at least a metre from anybody's face. And check that the window you intend to use actually opens.

Sensor settling time	**30 seconds** after power-up before the readings can be trusted, then a new value **about every five seconds**. **For the baseline, two to three minutes** of settling in a quiet room — and this is the shortcut taken most often and the most costly.
Sensors and boxes	SCD41. The CO_2 reading is essential; temperature and humidity are used as supporting evidence — both rise with it in a full room and both fall when it is aired. One box per room is enough; this activity is about *one* room, not about comparing boxes. **CO_2 needs recent firmware**: 2.2.0 or later. A box still on the old USB-only firmware is told by the app to flash the WiFi firmware, which works over USB too (`firmware/FLASHING_GUIDE.pdf`).

Three routes

Short	20–25 min: predictions (4), the baseline in a quiet, aired room (5), everyone comes in and the CO_2 climbs (6), the window opened and the recovery timed (5), conclusions (4). This is the wizard as written, with the predictions the app asks for too.
Full	The 50-minute lesson: the short route done carefully, with all four measurements written down and **the rates of rise and fall computed**. Export at the end.
Project	Project: lesson 1 as above; **one full logging day** with a paper log taped to the wall; lesson 2 opening the day-long export, matching every peak to an entry in the log and finding the worst hour; lesson 3 writing and presenting a ventilation recommendation for their own classroom.

A clean baseline costs five minutes — and without it you have nothing

The commonest way to spoil this activity is to start measuring in a room that thirty pupils have just walked into. The "baseline" is then not the empty room's but the full room's, and all the rise that follows looks far smaller than it is.
Air the room during the previous break, close the window, and **let the SCD41 settle for two to three minutes** before recording the baseline. If your timetable does not allow that, take the baseline *outdoors* or in an empty corridor and **say clearly to the class that this is what you have done** — a declared reference is a piece of data; an undeclared one is a hidden error.

1. Where this fits in the curriculum

- **Biology** — in **year VII**, human respiration is the most direct entry in the whole collection: the syllabus asks for the process, and here it has a measurable result, produced by the class itself. Every pupil in the room is a small, continuous CO_2 source, and the graph shows their sum.
- **Mathematics** — in **year VI**, ratio and proportion. **The rate of rise, in ppm per minute, is the ratio of two differences** and it is the most transferable thing pupils learn here: not "how big is the number" but "how fast is it changing". In years VII–VIII, interpreting a curve that flattens is added.
- **Geography** — in **year V**, the Earth's atmosphere and the composition of air; the outdoor value, around 420 ppm, becomes a measured number rather than a read one.
- **Physics** — in **year VI**, measuring physical quantities and the idea of an instrument; later, diffusion and the mixing of gases, if you want to explain why fresh air does not reach the whole volume instantly (check the year in the syllabus in force).
- **Personal development** — the activity ends with **a written recommendation, taken to the class teacher or the head**. It is one of the few occasions on which pupils produce evidence about their own environment and use it to ask for a change.

The core skill is **reading a rate**, not a value. A large number says little on its own; *how fast* it rises says how full and how unventilated the room is, and *how fast* it falls says how well the air is exchanged.

2. Time plan

The in-app wizard runs in about twenty minutes. This is, however, one of the strongest candidates in the whole collection for a genuine multi-day project: **a classroom's CO_2 curve over a full school day is a real dataset that says something true about the school**, and pupils can act on it.

Route	Time	What happens
Short	**20–25 min**	Predictions (4), the baseline in a quiet, aired room (5), everyone comes in and the CO_2 climbs (6), the window opened and the recovery timed (5), conclusions (4). This is the wizard.
Lesson 1	50 min	The short route done carefully, with all four measurements written down and the rates of rise and fall computed. Export at the end.
Between lessons	one school day	The box is left logging in the classroom from before the first lesson until after the last, with a simple **paper log**: who was in the room, and when the windows were opened. **No pupil time is used.**
Lesson 2	50 min	Open the day-long export, plot it, and match every peak and dip to an entry in the paper log. Identify the worst hour of the day.
Lesson 3	50 min	Pupils write and present a **ventilation recommendation** for their own classroom, with evidence, and take it to the class teacher or the head.

Total commitment for the full project: three 50-minute lessons (2 h 30 min) of class time spread over about a week, plus one unattended logging day. The short version fits comfortably inside a single lesson if that is all the timetable allows.

3. Materials

Item	Per group	Notes
MagicBox with SCD41	1	The CO_2 sensor is essential. It also reports temperature and humidity, which the activity uses as supporting evidence.
Laptop with the app	1	USB cable or WiFi; for CO_2 the box needs firmware 2.2.0 or later.
A room with a window that opens	1	The whole activity depends on being able to open and close it *deliberately*. A door to a corridor is a poor substitute but usable.
Stopwatch or phone timer	1	For timing the recovery. The app timestamps everything, but pupils should time it themselves as well.
Class register or a tally sheet	1	To record exactly how many people are in the room at each stage. **This number is data, not a detail.**
Sticky notes or masking tape	a few	To mark the sensor position so that it is identical on every measurement day.
Thermometer, or the box's own reading	1	Outdoor temperature matters: on a cold day ventilation is much faster and the room cools quickly.
Metre rule or tape measure	1	For the room's volume (length × width × height), without which the rate of rise cannot be predicted.
Extra items for the multi-day version		
Mains power or a power bank	1	A full school day of logging needs the box powered continuously.
A printed paper log taped to the wall	1	Columns for time, number of people present, window open or closed. Pupils fill it in as they arrive and leave.

Preparation the day before. Power up the box and leave the SCD41 running for at least ten minutes to check that its readings look sensible — outdoors or by an open window it should settle near 400–450 ppm. Decide and mark where the box will stand: roughly at the head height of a seated pupil, away from the window, away from the door, and at least a metre from anybody's face. Print the sheets, and check that the window you intend to use actually opens and that you are allowed to open it.

4. Scientific background for the teacher

Where the carbon dioxide comes from, and where it goes

Every person in the room is a small, continuous CO_2 source. A seated adult exhales roughly 300 mL of CO_2 per minute; a seated pupil somewhat less, about 150–200 mL/min. Exhaled breath is about 4–5% CO_2, that is 40 000–50 000 ppm, against an outdoor background of about 420 ppm.

A room is a well-mixed box of volume V with N occupants each producing CO_2 at rate G, with air exchanged with outside at Q cubic metres per second. The concentration c obeys a

simple mass balance:

$$V\frac{\mathrm{d}c}{\mathrm{d}t} = NG - Q\,(c - c_{\text{out}}).$$

Two things follow immediately, **and both are visible on the pupils' graph**.
The initial slope is set by people, not by ventilation. Right after the class enters, c is still close to c_{out}, so the loss term is almost zero and

$$\left.\frac{\mathrm{d}c}{\mathrm{d}t}\right|_{t=0} \approx \frac{NG}{V}.$$

The rise rate is proportional to the number of people and inversely proportional to the room volume. **The order of magnitude:** 30 pupils at 0.18 L/min each produce 5.4 L of CO_2 a minute; in a 170 m^3 room, that is 170 000 L, this means $5.4/170\,000 \approx 32 \times 10^{-6}$ per minute, so **about** 30 **ppm a minute, or roughly** +1000 **ppm in half an hour** with the room shut. **This is why the same class produces a much steeper line in a small room**, and it is a prediction pupils can test by moving to a different room.
The room settles at a steady state. If nothing changes, the rise flattens out at

$$c_\infty = c_{\text{out}} + \frac{NG}{Q},$$

which is exactly why measured CO_2 is used as a proxy for ventilation rate per person. A classroom that stabilises at 2000 ppm is not dangerous in itself; it is telling you that Q/N is small. (For older pupils: the steady state is reached after a few time constants $\tau = V/Q$, so in a well-sealed room only after hours — which is why, within one lesson, the curve is seen mostly rising.)

4.1 Why the fall is faster than the rise

When the window opens, Q increases by a large factor and the equation becomes an exponential decay towards c_{out}:

$$c(t) = c_{\text{out}} + \left(c_0 - c_{\text{out}}\right)\mathrm{e}^{-t/\tau}, \qquad \tau = \frac{V}{Q}.$$

The time constant τ is the air-change time of the room. With a single window ajar, τ might be twenty minutes; with a window and a door open on opposite sides, **cross-ventilation can bring it under five**.

This is the most useful practical result of the whole activity, and pupils can measure it: **the recovery is not linear**, it is steepest at the start and flattens as the room approaches outdoor air. The sensor itself has a response time of about a minute and one reading every five seconds, so a very fast fall, with cross-ventilation, appears slightly delayed and rounded on the graph; for a whole room, which changes over minutes, the effect is small.

4.2 Thresholds, and what they actually mean

The World Health Organization has no guideline value for indoor CO_2. The European reference for ventilation is the standard **EN 16798-1**, which places buildings in categories by how far indoor CO_2 exceeds the outdoor level: about +550 ppm (category I), +800 ppm (category II, the normal level for new buildings) and +1350 ppm (category III) — roughly 970, 1220 and 1770 ppm with outdoor air at 420 ppm. The app colour-codes the reading with simpler, rounded bands that say the same thing (on the chart the background bands carry the same names, *Excellent*, *Good*, *Moderate* and *Poor*; the verdict under the cards adds in brackets what to do, as in the table):

CO_2 level	The app's verdict	What it means
below 800 ppm	**Excellent**	Well ventilated; outdoor air is reaching the room.
800–1200 ppm	**Good**	Acceptable (around categories I–II of EN 16798-1), but the room would benefit from airing.
1200–2000 ppm	**Moderate (air the room soon)**	Insufficient ventilation: the air feels stuffy and many people feel drowsy.
2000–5000 ppm	**Poor (open the windows now)**	Clearly insufficient ventilation; complaints of tiredness and headaches are more frequent.
above 5000 ppm	**Dangerous (open the windows now)**	The 8-hour occupational exposure limit; not reachable in a normal classroom.

Be accurate with pupils: **CO_2 at classroom levels is not itself toxic**, and at 1500 ppm breathing is not "harder" — the room's oxygen falls by only a few tenths of a per cent. It is a **ventilation marker**: a high reading says that the air in the room has largely been through somebody's lungs already, and along with the CO_2 the moisture, smells and, in the cold season, viruses that people bring have built up too. Conclusions about specific pollutants — dust, smoke, volatile organic compounds — require **other measurements**, and this sensor does not see them.

5. Lesson flow

The in-app wizard has eight steps: the introduction, the prediction, the baseline in the aired room, filling the room, ventilation, analysis, the **How sure are we?** step and the conclusion. Before the measurements the app asks for a **written prediction**, keeps it in the group's journal and quotes it back in the conclusion. Below is what the teacher does around the steps.

1. **Written predictions, before anything (4 min).** Every pupil writes: *how much will the CO_2 rise in a quarter of an hour with the whole class in the room? And once we open the window, how long until it comes back to where it started?* Almost every class underestimates the rise and **overestimates** the speed of the recovery. The group's prediction also goes into the app's prediction box.
2. **The introduction on screen (3 min).** Read the thresholds together. The activity opens from the **Activities** page with **Start**, and the step ends with **Got it!**

Technical note — the baseline, the sensor's rhythm, and what gets recorded

The sensor. SCD41, an NDIR carbon dioxide sensor that also reports temperature and humidity. The screen shows three cards: **CO_2** in `ppm`, **Temperature** in °C and **Humidity** in per cent. In the exported files the columns are `CO2`, `T` and `H`. CO_2 is read by firmware 2.2.0 or later; if the app reports that a box runs the old USB-only firmware, reflash it with the WiFi one.

How long you wait. Thirty seconds after power-up, during which the readings are ignored; then a new value **about every five seconds**, so the graph advances in steps. **For the baseline, two to three minutes** of settling in a quiet room — and this is the shortcut that spoils the activity most often.

Where the box stands — and why it matters more here than anywhere else. The activity measures *the room*, not any one person's breathing. The box stands at the head height of a seated pupil, **at least a metre from anybody's face**, away from the window and away from the door. Mark the position with tape: if it moves between days, the multi-day series is no longer comparable.

What is recorded alongside the number. How many people are in the room, and **whether the window is open**. Without those two columns a day of data cannot be

interpreted at all — and with them, every peak and every dip acquires an explanation.
What the screen shows. Under the cards, on the left, the verdict **Air quality:** (for example *Good*); on the right, **Rate (last 3 min):**, the CO_2 gradient over the last three minutes, with an arrow — "↑ 24.0 ppm/min". At the baseline there is also a settling timer of 150 seconds ("Let the box settle: … s more"), which turns green when the wait is over; if you moved the box, the **I moved the box** button starts it again. The rates the analysis works with are the ones on screen at the moment you move to the next step, with their ±.
The buttons you will use. The green button changes its label at every step: **Got it!**, **My prediction is written**, **I noted the baseline values**, **I watched the CO_2 rise**, **I aired out the room**, **I analyzed the data**, **Continue**, and finally the blue **Finish** button. At the three measurements, **Capture** (under the chart) keeps the CO_2, temperature and humidity: at the baseline, after the settling; while filling the room, at the highest value; during ventilation, when you close the window. Above the chart sit **Smooth: on/off**, **Pause** and **Clear**. The chart starts at 0:00 at every step, with time in minutes and seconds — exactly the axis the rates need.
Exporting the data. At the measurement steps, **PNG** (all the step's charts) and **Excel** (the step's raw, unfiltered readings, with an `elapsed_s` column of seconds elapsed) appear. Predictions and notes are saved in the group's journal, in the `logs` folder, and as a **Journal** sheet in the activity's Excel file. **There is no CSV button inside the activity**; files for the whole day come from the **Data & Export** page (as `.xlsx` by default). The app no longer deletes old logs by itself. For the full logging day, export **at the end of the day**, before closing the app, and save following the pattern `class_date_activity`.

3. **The baseline, in a quiet aired room (5 min).** With the room aired during the break and the window closed, the box is left for two to three minutes and the CO_2, temperature and humidity are recorded. **Record how many people are in the room at that moment too.**

4. **Filling the room (6 min).** Everyone comes in and sits down. Watch the line rise. Record the value every five minutes — this is when the data for *the rate of rise* is gathered. Ask: *why is it rising faster now than it will in twenty minutes?*

5. **Ventilation (5 min).** Open the window — and, if you can, the door on the opposite side — and start the stopwatch. Record the value every minute. **Here it becomes visible that the recovery is not linear**: it falls steeply at first and more and more slowly after.

6. **Analysis and conclusion (4 min).** The app's analysis step gathers the group's data — the baseline, the highest value, the rise rate and the fall rate, with ± — and asks for three calculations in the journal: the rate per person, the build-up factor (highest value divided by the baseline) and which was faster. The **How sure are we?** step reminds them that the sensor is off by about $\pm(40\ \text{ppm} + 5\%)$ as a value but measures changes far better, that two boxes are compared only after standing side by side, and that a rate is real only if it is clearly bigger than its ±. State the class's conclusion aloud: our room rises by so many ppm a minute when we are all inside, and falls by so many when we open the window. **From there comes the recommendation:** how many times an hour, and for how long?

The investigation, in one table

The question. How fast does the air in our room go stale, and how fast does it come right again?
The hypothesis. Written **before** any measuring, as two numbers: how much the CO_2 will rise in a quarter of an hour, and in how many minutes it will come back after the window is opened. Not changed afterwards.
What I change. One thing at a time: **the number of people** in the room, then **the**

state of the window.
What I measure. CO_2, in ppm, **and the time**, at fixed intervals — every five minutes during the rise, every minute during the recovery. Without recorded times there are no rates.
What I keep the same. The same box, in the same marked position, at the same height; nobody closer than a metre to it; the same door, open or closed, for as long as the rise lasts.
How many repeats. The rise can be repeated the next day with a different number of pupils — **and that is the best possible repeat**, because it tests the prediction that the rate is proportional to the number of people.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken with the box in its marked place, with the number of people recorded and the state of the window recorded. An artefact is anything else: **a pupil leaning over the box to see the screen produces a sharp peak that has nothing to do with the room**; a door left open by accident flattens the whole rise; and a baseline taken after the class has come in makes the entire rise look half as big as it is.

6. Analysis after the activity

6.1 What the data should look like

Situation	Typical value	What it means
Outdoors	400–450 ppm	The atmospheric background; the floor below which you cannot go indoors.
Empty, aired room	450–700 ppm	A good baseline.
Room with 30 pupils, after 15 min	800–1200 ppm	Close to a linear rise at first.
The same room, after 45 closed minutes	1300–2000 ppm	The curve begins to flatten, the sooner the more air the room lets in.
Rate of rise, 30 pupils, ≈ 170 m^3	20–40 ppm/min	Proportional to the number of people, inversely proportional to the room volume: about +1000 ppm in 30 min.
Rate of fall, one window	40–100 ppm/min at first	Slows as the room approaches outdoor air.
Rate of fall, window and door	150–400 ppm/min at first	Cross-ventilation; several times faster.
Temperature and humidity, on airing	both fall	The supporting evidence: outdoor air came in, not merely less CO_2.

Absolute values depend on the number of pupils, the room volume, how well the windows seal and the weather. **Do not look for a "correct" number.** What must come out reliably is the shape: close to a linear rise at first, a slow flattening, then a steep fall that eases off.

6.2 From two numbers to a rate, step by step

This is the central operation of the activity, and in year VI it is done in full. Three steps, on the board:

1. **The difference in concentration.** $1150 - 700 = 450$ ppm.
2. **The difference in time.** From 9:05 to 9:20 is 15 minutes. **This is why the time is**

recorded alongside the number, and why a sheet with no time column cannot produce a rate.

3. **The ratio.** 450 : 15 = 30 ppm per minute. Say it in words: *the air in our room loads up by about thirty ppm every minute that we all sit in it with the window shut.* Check it against the calculation in the scientific background: 30 pupils in a 170 m^3 room also give about 30 ppm/min.

Then ask the question that leads to the recommendation: *if we rise at* 30 *ppm a minute and fall at* 200 *when the window is open, how often and for how long must we air the room to stay below* 1200*?* (The simplified answer: from 700 to 1200 takes about 17 minutes, so the airing has to come about every quarter of an hour, or the window must stay ajar.) It is a proportionality problem with an answer the class can put into practice tomorrow.

6.3 Questions to ask afterwards

1. The rise was steeper in the first five minutes than in the last five. Why does it ease off, when there are just as many of us in the room?
2. The same class in a room twice the size: what would happen to the rate of rise? Justify your answer.
3. When we opened the window, the temperature and the humidity fell as well as the CO_2. What does that tell us about what actually came into the room?
4. The outdoor value was 420 ppm. Why can we not go below it, however much we air the room?
5. Our room stayed below 5000 ppm throughout. Does that mean the air was good? What does the standard EN 16798-1 say about our value? What are we actually measuring with this sensor — and what are we *not* measuring?

6.4 Assessment ideas

- **Low stakes:** the pupil completes the four values and says which band the room was in at each stage.
- **Standard:** the pupil computes the rate of rise and the rate of fall correctly, in ppm/min, and writes a sentence comparing the two.
- **Extension: the ventilation recommendation**, at most one page — the problem, two pieces of evidence from their own data, the proposed intervention (how often per hour and for how long), the expected result, and **a stated limitation of the measurement**. That last is what separates a very good piece of work from a good one.

Common pitfalls and misconceptions

The baseline taken after the class has come in. The number one mistake, and the most damaging: if the "empty room" already had thirty pupils in it, the whole subsequent rise looks half as big. Air the room during the break, close the window, wait two to three minutes.
Pupils leaning over the box. They produce sharp peaks that have nothing to do with the room. A metre away, screen projected or a named reader.
Recording the number without the time. Without times there are no rates, and the rate is the lesson. The time column is not optional.
Forgetting the number of people and the state of the window. A whole day of data without those two columns cannot be interpreted at all. The paper log on the wall exists for exactly this.
Expecting a linear recovery. The recovery is exponential: fast at first, slow at the end.

Pupils who measure only the start and the end get an "average rate" that describes neither phase.

"At high CO_2 it is harder to breathe." No: the oxygen falls by only a few tenths of a per cent. The feeling of stuffy air comes from the warmth, moisture and smells that build up along with the CO_2.

"High CO_2 means polluted air." It means poor ventilation. **The sensor does not see dust, smoke or smells**, and a conclusion about specific pollutants needs other instruments. Say so explicitly — it is precisely the limitation the written recommendation calls for.

Moving the box between days. It makes the multi-day series incomparable. Mark the position with tape.

The door left open during the rise. It flattens the curve and makes the room look far better ventilated than it is. Fix the state of the door at the start and do not change it.

Safety and classroom management

Windows are opened by the teacher, not by pupils, and nobody climbs on a sill, a radiator or a chair to reach the handle. If the window opens onto a yard with people in it, check before opening.

Cold. On a cold day, airing chills the room quickly and the pupils by the window freeze. **Air briefly and fully** — the window wide open for a few minutes — rather than a little for a long time, and move the pupils away from the window while it is open. This is, incidentally, also the correct recommendation from an energy point of view.

Draughts. A window and a door open on opposite sides produce a strong draught: papers fly and the door can slam on fingers. Wedge the door and gather the papers first.

Cables. A box left logging all day is mains-powered: the cable does not cross the walkway, and the box stands somewhere it cannot be knocked over.

Equipment left unattended. A box left logging through the day is announced to colleagues who use the room and labelled with the class and teacher name, so that it is not moved or switched off.

Adaptations for pupils with special educational needs

The table with the times pre-filled. This is the main adaptation and it is not a concession: the sheet arrives with the time column already written — 0, 5, 10, 15 minutes — and the pupil fills in only the values. **The arithmetic of time is not the skill being assessed here**; the ratio is.

One rate, worked in full. The first rate calculation comes printed on the sheet, with all three steps, as a pattern to follow. The support is in the model, not in the answer.

Without computing rates at all. For year V, or for pupils who do not yet work with ratios, the whole activity can be done by comparing *shapes*: where the line is steep and where it is gentle. The pupil circles the word from the screen at each stage and draws the shape of the curve. The conclusion — the room goes stale slowly and comes right quickly when we open the window — is reached in full.

Three real roles for pupils with motor difficulties. The activity is static anyway, but the roles make participation visible. They rotate at every stage:

- **the timekeeper** — announces the reading moments, every five minutes during the rise and every minute during the recovery; this is the role with the greatest scientific value, because without times there are no rates;
- **the counter** — keeps track of how many people are in the room and announces any change, including when somebody steps into the corridor;
- **the screen reader** — says the three values aloud.

Sensitivity to cold and draughts. Pupils sitting by the window move during the airing.

Announce in advance how long it will last, so that nobody is taken by surprise.

If things do not go to plan

You cannot air the room during the previous break. Take the baseline **outdoors** or in an empty corridor and **tell the class explicitly that you have done so**. A declared reference is data; an undeclared one is a hidden error. The rise remains perfectly measurable.
The window will not open. Use the door to the corridor. The recovery will be much slower, and that is itself good evidence for the written recommendation: *our room can only be aired through the door, and that is why it needs longer.*
You have no full logging day. The short route has all four measurements and both rates. The full day adds richness, not the conclusion.
The value does not rise at all, although the room is full. Check whether the door or a window has been left ajar — that is the usual cause. If they are shut and it still does not move, check that the thirty seconds since power-up have passed and that the chart is not on **Pause**.
The value is already above 2000 **ppm at the start of the lesson.** Not a problem but the discovery of the day. Begin by airing, measure the recovery, then close up and measure the rise. The order changes, the content does not — **and the written recommendation becomes far more convincing**.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils compute both rates from the given values, compare them and write the recommendation. **The calculation and the decision, which are the lesson, are done entirely without the equipment.** Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **World Health Organization** — `https://www.who.int` — search for indoor air quality; its guidelines cover pollutants (particles, combustion gases, damp), **not CO_2**, for which the WHO gives no guideline value.
- **REHVA** — `https://www.rehva.eu` — the European federation of building services associations; search for the standard EN 16798-1 and their guidance on classroom ventilation, if you want professional figures for the pupils' recommendation.
- **Institute of Physics** — `https://www.iop.org` — search for material on gases and diffusion, for the mechanism.
- **STEM Learning** — `https://www.stem.org.uk` — search for respiration and air quality for ages 11–14.
- **NASA Climate** — `https://climate.nasa.gov` — for the link between the room's CO_2 and the atmosphere's, if a pupil asks.
- **Romanian Ministry of Education** — `https://www.edu.ro` — for the norms concerning classrooms, if the pupils' recommendation reaches the school leadership and needs an official framing.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *how many times an hour should our room be aired, and how do you know?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer level the three axes look like this:

- **Measurement practice** — the pupil takes the baseline in an aired room, records the time, the number of people and the state of the window alongside the value, and stays back from the sensor.
- **Data interpretation** — the pupil computes the rates correctly in ppm/min, notices that the rise flattens and the recovery eases off, and does not confuse a large value with a large rate.
- **Scientific reasoning** — the pupil explains where the CO_2 comes from, frames a ventilation recommendation supported by their own data, and **states a limitation** — that they are measuring ventilation, not pollutants.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the pupils fill in is a separate document, in the folder `docs/en/student_sheets`: the file `indoor_pollution.pdf`. It has two pages — the predictions and the table of the rise on the first; the table of the recovery, the rate calculations and the recommendation on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The time column is already printed** — 0, 5, 10, 15 minutes for the rise and every minute for the recovery — because without times there are no rates, and the arithmetic of time is not the skill being assessed here. **The first rate calculation comes worked in full**, with all three steps, as a pattern to follow; the others are not pre-computed, because the ratio *is* the lesson. **The sheet asks for the number of people and the state of the window** at each stage, in boxes of their own: without them the data cannot be interpreted an hour later, let alone a week. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.